

K-MEANS RESULTS AND INTERPRETATION

The K-Means algorithm with K=3 resulted in a well-defined and interpretable clustering of the student population. When the three clusters were separated, they did so mainly in the dimensions Attendance and Hours_Studied, which were the two top dimensions of the ranking of the J48 InfoGain.

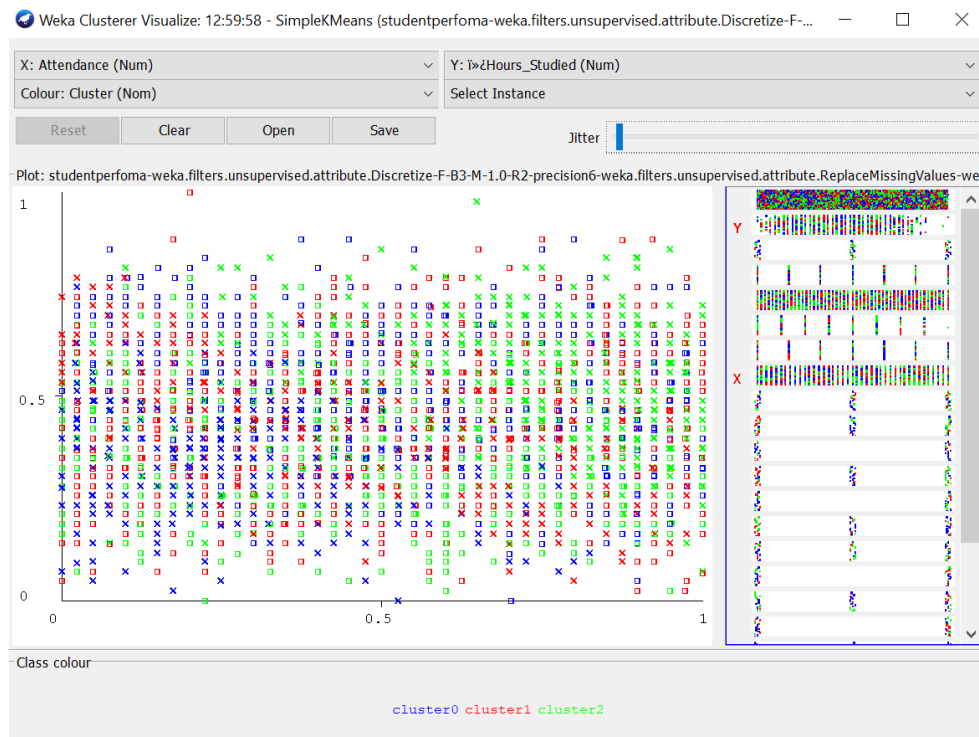
Final cluster centroids:

Attribute	Cluster#			
	Full Data (6607.0)	0 (2093.0)	1 (2106.0)	2 (2408.0)
Hours_Studied	0.4413	0.4432	0.4427	0.4384
Sleep_Hours	0.5048	0.5114	0.5024	0.5013
Previous_Scores	0.5014	0.4748	0.5691	0.4653
Tutoring_Sessions	0.1867	0.1859	0.1861	0.188
Physical_Activity	0.4946	0.4982	0.4877	0.4976
Attendance	0.4994	0.4266	0.5046	0.5582
Parental_Involvement	Medium	Medium	Medium	Medium
Access_to_Resources	Medium	Medium	High	Medium
Extracurricular_Activities	Yes	Yes	Yes	Yes
Motivation_Level	Medium	Medium	Low	Medium
Internet_Access	Yes	Yes	Yes	Yes
Family_Income	Low	Medium	Medium	Low
Teacher_Quality	Medium	Medium	Medium	High
School_Type	Public	Public	Public	Public
Peer_Influence	Positive	Negative	Neutral	Positive
Learning_Disabilities	No	No	No	No
Parental_Education_Level	High School	College	High School	High School
Distance_from_Home	Near	Near	Near	Near
Gender	Male	Male	Female	Male

Figure 8 : The image shows the K-Means Cluster Centroids for the Normalised Values with 3 Clusters.

SimpleKMeans clustered the students into three groups with 2,093 students (32%) in Cluster 0, 2,106 students (32%) in Cluster 1, and 2,408 students (36%) in Cluster 2. If compared to the Exam_Score performance classes, Cluster 0 is classified as the Low performers, Cluster 1 as the Medium performers and Cluster 2 as the High performers. The most informative separator across the clusters is the Attendance centroid, which is 0.4266 for Cluster 0 (Low), 0.5046 for Cluster 1 (Medium) and 0.5582 for Cluster 2 (High). This gradient supports the idea that in unsupervised analysis, attendance is always a significant factor for performance grouping, directly supporting the classification tree, J48, which has Attendance as the root node (Quinlan, 1993).

In addition to attendance, Previous_Scores also meaningfully differentiates the clusters; Cluster 1 (Medium) has the highest normalised centroid (0.5691), while Cluster 0 (Low) has a normalised centroid of 0.4748 and Cluster 2 (High) has a normalised centroid of 0.4653, indicating that an



individual's prior academic history is a nuanced factor. Interestingly, there are also differences in the categorical centroids: Cluster 0 is characterized by a Negative Peer_Influence centroid, Cluster 1 by a Neutral Peer_Influence centroid and a High Access_to_Resources centroid and a Low Motivation_Level centroid, and Cluster 2 by a Positive Peer_Influence centroid, a High Teacher_Quality centroid and a Low Family_Income centroid. 4,158 instances (62.93%) were incorrectly clustered, which is not surprising, since clustering of the mixed-type data with 20 attributes is a much more difficult problem than supervised classification.

Figure 9 : WEKA SimpleKMeans cluster visualisation (Attendance vs Hours_Studied).

Three clusters visible. Cluster 0 (2,093 students, 32%) maps to Low performers. Cluster 1 (2,106 students, 32%) maps to Medium performers. Cluster 2 (2,408 students, 36%